

# DAY 4 TASK

## What is an API ?

“API stands for application programming interface . it is a way for two software applications to talk with each other “.

- The client asks what you want API takes the information from server just like bride.
- **Request** : what client ask(query)
- **Response** : what API gives to client from server

**Methods** — The type of action you want to perform:

- GET — Retrieve data
- POST — Send/create data
- PUT / PATCH — Update data
- DELETE — Remove data

## Types of APIs

- **REST** — Most common; uses HTTP methods (what most web APIs use)
- **GraphQL** — Flexible querying; you ask for exactly what you need
- **SOAP** — Older, XML-based; used in enterprise systems
- **WebSocket** — Real-time, two-way communication

## What is an AI API ?

“An AI API is an interface that lets you access AI models and capabilities through simple code ,without building the AI yourself. You send a request, and the AI processes it and sends back a response.

## How it works:

**Your App → [Request] → AI API → AI Model → [Response] → Your App**

## What is an Gemini API ?

“The Gemini API is Google's AI API that gives you access to their Gemini family of AI models. It allows developers to build apps with powerful AI capabilities like text, image, audio, and video understanding.”

## How to Get Started

**Step 1** — Go to [Google AI Studio](#)

**Step 2** — Create an API Key (free to start)

**Step 3** — Install the SDK

bash

```
npm install @google/generative-ai
```

**Step 4** — Make your first API call

## Difference between normal python program & AI API chatbot :

### Normal python program :

This normal python follows fixed rules written by the programmer

You: tell a joke → Bot: I don't understand that

### **AI API Chatbot :**

An AI API chatbot uses a trained AI model that understands language naturally and can answer almost anything.

You: tell a joke → Bot: Why did the developer quit? No more exceptions